Solutions

Maths

Q1. Let $A=\begin{bmatrix}1 & 1 & 1\\ 0 & 1 & 1\\ 0 & 0 & 1\end{bmatrix}$. If A^n represents the n-th power of matrix A for $n\in\mathbb{N}$, and its inverse $(A^n)^{-1}$ is obtained by applying elementary row operations, then the sum of all elements of the matrix $(A^{10})^{-1}$ is equal to:

(A) 18 (B) 38 (C) 28 (D) 48

Ans. C

Sol. $A^2=\begin{bmatrix}1 & 1 & 1\\ 0 & 1 & 1\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 1 & 1\\ 0 & 1 & 1\\ 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}1 & 2 & 3\\ 0 & 1 & 2\\ 0 & 0 & 1\end{bmatrix}$

$A^3=\begin{bmatrix}1 & 3 & 6\\ 0 & 1 & 3\\ 0 & 0 & 1\end{bmatrix}$

$(A^n)^{-1}=\begin{bmatrix}1 & -n & \frac{n(n-1)}{2}\\ 0 & 1 & -n\\ 0 & 0 & 1\end{bmatrix}$.

For $n=10$: $(A^{10})^{-1}=\begin{bmatrix}1 & -10 & 45\\ 0 & 1 & -10\\ 0 & 0 & 1\end{bmatrix}$

Sum of all elements $=1-10+45+0+1-10+0+0+1=28$.

Q2. The sum of all solutions of the trigonometric equation $\sqrt{3}\sin 2x+\cos 2x-\sqrt{3}\sin x-3\cos x+2=0$ in the interval $[0,2\pi]$ is equal to

(A) $\frac{11\pi}{3}$ (B) $\frac{14\pi}{3}$ (C) 4π (D) $\frac{13\pi}{3}$

Ans. B

Sol. $\sqrt{3}\sin 2x+\cos 2x-\sqrt{3}\sin x-3\cos x+2=0$

$2\sqrt{3}\sin x\cos x+(2\cos^2 x-1)-\sqrt{3}\sin x-3\cos x+2=0$

$\Rightarrow (2\cos x-1)(\sqrt{3}\sin x+\cos x-1)=0$

Case 1 :

$2\cos x-1=0 \implies \cos x=\frac{1}{2}$

In $[0,2\pi]$, $x=\frac{\pi}{3},\frac{5\pi}{3}$.

Case 2 :

$\sqrt{3}\sin x + \cos x - 1 = 0$

$\implies \dfrac{\sqrt{3}}{2}\sin x + \dfrac{1}{2}\cos x = \dfrac{1}{2}$

$\Rightarrow \cos\left(x - \dfrac{\pi}{3}\right) = \dfrac{1}{2}$

$\Rightarrow x - \dfrac{\pi}{3} = 2k\pi \pm \dfrac{\pi}{3}, \quad k \in \mathbb{Z}$

For $k = 0$: $x = 0$ or $x = \dfrac{2\pi}{3}$.

For $k = 1$: $x = 2\pi$ or $x = \dfrac{8\pi}{3}$ (outside $[0, 2\pi]$).

Thus, the distinct solutions in $[0, 2\pi]$ are $x \in \left\{0, \dfrac{\pi}{3}, \dfrac{2\pi}{3}, \dfrac{5\pi}{3}, 2\pi\right\}$.

Sum of all solutions $= 0 + \dfrac{\pi}{3} + \dfrac{2\pi}{3} + \dfrac{5\pi}{3} + 2\pi = \dfrac{14\pi}{3}$.

Q3. Let x_1 and x_2 be the real roots of the logarithmic equation $(\log_2 x)^2 - 4\log_2 x + 2 = 0$ If $\alpha = \log_{x_1}(4x_1)$ and $\beta = \log_{x_2}(4x_2)$, then the quadratic equation with leading coefficient 1 whose roots are α and β is:

(A) $x^2 - 6x + 7 = 0$ (B) $x^2 - 6x + 5 = 0$ (C) $x^2 - 4x + 7 = 0$ (D) $x^2 - 8x + 7 = 0$

Ans. A

Sol. Let $t = \log_2 x$.

$t^2 - 4t + 2 = 0$

$\alpha = \log_{x_1}(4x_1) = \log_{x_1} 4 + \log_{x_1} x_1$

$= \dfrac{2}{\log_2 x_1} + 1 = 1 + \dfrac{2}{t_1}$

$\beta = \log_{x_2}(4x_2) = \log_{x_2} 4 + \log_{x_2} x_2$

$= \dfrac{2}{\log_2 x_2} + 1 = 1 + \dfrac{2}{t_2}$

$y = 1 + \dfrac{2}{t} \implies t = \dfrac{2}{y-1}$

Substituting $t = \dfrac{2}{y-1}$ into $t^2 - 4t + 2 = 0$:

$\left(\dfrac{2}{y-1}\right)^2 - 4\left(\dfrac{2}{y-1}\right) + 2 = 0$

$\dfrac{2}{(y-1)^2} - \dfrac{4}{y-1} + 1 = 0$

$y^2 - 6y + 7 = 0$

Thus, the required quadratic equation in x is $x^2 - 6x + 7 = 0$.

Q4. For each integer $k \geq 1$, let a_k be defined as $a_k = \dfrac{2k + 1 - 2\sqrt{k^2 + k}}{k^2 + k}$. If $x_1, x_2, \ldots, x_{35}$ are positive real numbers such that $\displaystyle\sum_{k=1}^{35} x_k = 50$, then the minimum possible value of the sum

$\sum_{k=1}^{35} \frac{a_k}{x_k}$ is

(A) $\frac{1}{36}$ (B) $\frac{1}{72}$ (C) $\frac{25}{72}$ (D) $\frac{5}{36}$

Ans. B

Sol. $a_k = \frac{(k+1)+k-2\sqrt{k(k+1)}}{k(k+1)}$

$= \left(\frac{\sqrt{k+1}-\sqrt{k}}{\sqrt{k(k+1)}}\right)^2$

$\sqrt{a_k} = \frac{1}{\sqrt{k}} - \frac{1}{\sqrt{k+1}}$

$\sum_{k=1}^{35} \sqrt{a_k} = \sum_{k=1}^{35} \left(\frac{1}{\sqrt{k}} - \frac{1}{\sqrt{k+1}}\right)$

$= 1 - \frac{1}{\sqrt{36}} = 1 - \frac{1}{6} = \frac{5}{6}$

$\sum_{k=1}^{35} x_k = 50$ and $\sum_{k=1}^{35} \sqrt{a_k} = \frac{5}{6}$:

$50 \cdot \sum_{k=1}^{35} \frac{a_k}{x_k} \geq \left(\frac{5}{6}\right)^2 = \frac{25}{36}$

$\sum_{k=1}^{35} \frac{a_k}{x_k} \geq \frac{25}{36 \times 50} = \frac{1}{72}$

Q5. Let $x_1 = 4$ and $y_1 = 1$. For all integers $n \geq 1$, the sequences (x_n) and (y_n) are defined by the recurrence relations:

$x_{n+1} = \frac{x_n + y_n}{2}$ and $y_{n+1} = \frac{2x_n y_n}{x_n + y_n}$

Which of the following statement(s) is/are **CORRECT**?

(A) $x_n y_n = 4$ for all $n \geq 1$. (B) $x_n = 2\left(\frac{3^{2^{n-1}}+1}{3^{2^{n-1}}-1}\right)$ for all $n \geq 1$.

(C) $x_n - y_n = \frac{8 \cdot 3^{2^{n-1}}}{9^{2^{n-1}}-1}$ for all $n \geq 1$. (D) $\prod_{k=1}^{n}\left(\frac{x_k+2}{x_k-2}\right) = 3^{2^n-1}$ for all $n \geq 1$.

Ans. A, B, C, D

Sol. **For Option (A):**

$x_{n+1}y_{n+1} = \left(\frac{x_n+y_n}{2}\right)\left(\frac{2x_ny_n}{x_n+y_n}\right) = x_ny_n$

$x_ny_n = x_1y_1 = 4 \times 1 = 4$ for all $n \geq 1$.

Thus, Option (A) is correct.

For Option (B):

$x_{n+1} = \frac{x_n + \frac{4}{x_n}}{2} = \frac{x_n^2+4}{2x_n}$

$x_{n+1} - 2 = \frac{(x_n-2)^2}{2x_n}$ and $x_{n+1} + 2 = \frac{(x_n+2)^2}{2x_n}$

Dividing the two equations yields:

$$\frac{x_{n+1}-2}{x_{n+1}+2}=\left(\frac{x_n-2}{x_n+2}\right)^2$$

Let $u_n=\dfrac{x_n-2}{x_n+2}$. Then $u_{n+1}=u_n^2$, which implies $u_n=u_1^{2^{n-1}}$.

Since $u_1=\dfrac{4-2}{4+2}=\dfrac{1}{3}$, we get:

$$\frac{x_n-2}{x_n+2}=\left(\frac{1}{3}\right)^{2^{n-1}}=3^{-2^{n-1}}$$

$$x_n=2\left(\frac{1+3^{-2^{n-1}}}{1-3^{-2^{n-1}}}\right)=2\left(\frac{3^{2^{n-1}}+1}{3^{2^{n-1}}-1}\right)$$

Thus, Option (B) is correct.

For Option (C):

Since $y_n=\dfrac{4}{x_n}$, we have:

$$x_n-y_n=x_n-\frac{4}{x_n}=2\left(\frac{3^{2^{n-1}}+1}{3^{2^{n-1}}-1}\right)-\frac{2\left(3^{2^{n-1}}-1\right)}{3^{2^{n-1}}+1}$$

$$x_n-y_n=2\left[\frac{\left(3^{2^{n-1}}+1\right)^2-\left(3^{2^{n-1}}-1\right)^2}{\left(3^{2^{n-1}}-1\right)\left(3^{2^{n-1}}+1\right)}\right]$$

$$=2\left(\frac{4\cdot 3^{2^{n-1}}}{3^{2^n}-1}\right)=\frac{8\cdot 3^{2^{n-1}}}{9^{2^{n-1}}-1}$$

Thus, Option (C) is correct.

For Option (D):

From Option (B), we have $\dfrac{x_k+2}{x_k-2}=3^{2^{k-1}}$.

Therefore:

$$\prod_{k=1}^{n}\left(\frac{x_k+2}{x_k-2}\right)=\prod_{k=1}^{n}3^{2^{k-1}}=3^{\sum_{k=1}^{n}2^{k-1}}=3^{2^n-1}$$

Thus, Option (D) is correct.

Q6. Let $\langle a_n\rangle_{n\geq 1}$ be a sequence of positive real numbers such that $a_1=3$, and the sequence of its consecutive differences $d_n=a_{n+1}-a_n$ forms an arithmetic progression with first term 5 and common difference 2. Define $b_n=\dfrac{1}{a_n a_{n+1}}$ for all $n\geq 1$.

Which of the following statement(s) is/are **CORRECT**?

(A) $a_n=n(n+2)$ for all $n\geq 1$. (B) $\displaystyle\sum_{n=1}^{\infty}b_n=\frac{1}{18}$.

(C) $\displaystyle\sum_{n=1}^{N}(a_{n+1}-a_n)b_n=\frac{N(N+2)}{3(N+1)(N+3)}$ for any positive integer N.

(D) $\displaystyle\sum_{n=1}^{\infty}\frac{n+2}{(n+4)a_n a_{n+1}}=\frac{5}{144}$.

Ans. A, B, D

Sol. $d_n = 5 + (n-1)2 = 2n + 3.$

$$a_n = a_1 + \sum_{k=1}^{n-1} d_k = 3 + \sum_{k=1}^{n-1}(2k+3)$$

$$= 3 + 2 \cdot \frac{(n-1)n}{2} + 3(n-1)$$

$$= n^2 + 2n = n(n+2)$$

Option (A) is correct.

$$b_n = \frac{1}{a_n a_{n+1}} = \frac{1}{n(n+2)(n+1)(n+3)}$$

$$= \frac{1}{n(n+1)(n+2)(n+3)}.$$

For Option (B):

$$b_n = \frac{1}{3}\left[\frac{1}{n(n+1)(n+2)} - \frac{1}{(n+1)(n+2)(n+3)}\right]$$

$$\sum_{n=1}^{\infty} b_n = \frac{1}{3}\left(\frac{1}{1 \cdot 2 \cdot 3}\right) = \frac{1}{18}$$

Option (B) is correct.

For Option (C):

$$(a_{n+1} - a_n)b_n = \frac{a_{n+1} - a_n}{a_n a_{n+1}} = \frac{1}{a_n} - \frac{1}{a_{n+1}}$$

$$\sum_{n=1}^{N}(a_{n+1} - a_n)b_n = \frac{1}{a_1} - \frac{1}{a_{N+1}}$$

$$= \frac{1}{3} - \frac{1}{(N+1)(N+3)} = \frac{N(N+4)}{3(N+1)(N+3)}$$

Since $\frac{N(N+4)}{3(N+1)(N+3)} \neq \frac{N(N+2)}{3(N+1)(N+3)}$, Option (C) is incorrect.

For Option (D):

$$\frac{n+2}{(n+4)a_n a_{n+1}} = \frac{n+2}{n(n+1)(n+2)(n+3)(n+4)}$$

$$= \frac{1}{n(n+1)(n+3)(n+4)}$$

Multiplying numerator and denominator by $(n+2)$:

$$T_n = \frac{n+2}{n(n+1)(n+2)(n+3)(n+4)}$$

$$= \frac{(n+4)-2}{n(n+1)(n+2)(n+3)(n+4)}$$

$$T_n = \frac{1}{n(n+1)(n+2)(n+3)} - \frac{2}{n(n+1)(n+2)(n+3)(n+4)}$$

$$\sum_{n=1}^{\infty} T_n = \frac{1}{18} - 2 \cdot \frac{1}{4 \cdot (1 \cdot 2 \cdot 3 \cdot 4)}$$

$$= \frac{1}{18} - \frac{1}{48} = \frac{5}{144}$$

Hence, Option (D) is correct.

Q7. Consider the system of linear equations in real variables x, y, z:

$x + y + z = 1$

$x + 2y + az = b$

$x + 4y + a^2z = b^2$

where $a, b \in \mathbb{R}$. Let A be the coefficient matrix of the system, and let $\Delta, \Delta_x, \Delta_y, \Delta_z$ denote the determinants defined according to Cramer's rule. Which of the following statements is/are **CORRECT**?

(A) For any $a \in \mathbb{R} - \{1, 2\}$ and $b \in \mathbb{R}$, the system has a unique solution with $z = \dfrac{(b-1)(b-2)}{(a-1)(a-2)}$.

(B) The third row of the coefficient matrix A is a linear combination of its first two rows if and only if $a \in \{1, 2\}$.

(C) The system of equations has infinitely many solutions if and only if $a \in \{1, 2\}$ and $b \in \{1, 2\}$.

(D) If $a = 2$ and $b = 3$, then $\Delta_x = 0$, but the system of equations is inconsistent.

Ans. A, B, C, D

Sol. $$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & a \\ 1 & 4 & a^2 \end{vmatrix} = (2-1)(a-1)(a-2)$$

$= (a-1)(a-2)$

For R_3 to be a linear combination of R_1 and R_2, let $R_3 = c_1R_1 + c_2R_2$:

$(1, 4, a^2) = c_1(1, 1, 1) + c_2(1, 2, a)$

$= (c_1 + c_2, c_1 + 2c_2, c_1 + ac_2)$

Comparing components:

$c_1 + c_2 = 1$ and $c_1 + 2c_2 = 4 \implies c_2 = 3, c_1 = -2$.

Thus, $R_3 = 3R_2 - 2R_1 \implies a^2 = 3a - 2$.

$\implies a^2 - 3a + 2 = 0 \implies a \in \{1, 2\}$

Hence, statement (B) is correct.

$$\Delta_x = \begin{vmatrix} 1 & 1 & 1 \\ b & 2 & a \\ b^2 & 4 & a^2 \end{vmatrix} = (2-b)(a-b)(a-2)$$

$$\Delta_y = \begin{vmatrix} 1 & 1 & 1 \\ 1 & b & a \\ 1 & b^2 & a^2 \end{vmatrix} = (b-1)(a-1)(a-b)$$

$$\Delta_z = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & b \\ 1 & 4 & b^2 \end{vmatrix} = (b-1)(b-2)$$

For $a \in \mathbb{R} - \{1, 2\}$, $\Delta \neq 0$, giving a unique solution with:

$$z = \frac{\Delta_z}{\Delta} = \frac{(b-1)(b-2)}{(a-1)(a-2)}$$

Hence, statement (A) is correct.

$b^2 = 3b - 2 \implies b^2 - 3b + 2 = 0 \implies b \in \{1, 2\}$

$a \in \{1, 2\}$ and $b \in \{1, 2\}$.

statement (C) is correct.

For $a = 2$ and $b = 3$:

$\Delta = (2 - 1)(2 - 2) = 0.$

$\Delta_x = (2 - 3)(2 - 3)(2 - 2) = 0.$

$\Delta_z = (3 - 1)(3 - 2) = 2 \neq 0.$

Since $\Delta = 0$ and $\Delta_z \neq 0$, the system has no solution

Hence, statement (D) is correct.

Q8. Consider the system of homogeneous linear equations in x, y, z for $\theta \in [0, 2\pi]$:

$x + (\sin\theta)y + (\cos\theta)z = 0$

$x + (\cos\theta)y + (\sin\theta)z = 0$

$x + y + z = 0$

Which of the following statement(s) is/are **CORRECT**?

(A) The system has non-trivial solutions for exactly two values of θ in $[0, 2\pi]$.

(B) Whenever the system admits non-trivial solutions, $x = 0$ for all such solutions.

(C) For $\theta = \dfrac{\pi}{2}$, the system has a unique solution.

(D) For $\theta = \dfrac{3\pi}{4}$, the system has infinitely many solutions.

Ans. A, B, C

Sol. $D(\theta) = \begin{vmatrix} 1 & \sin\theta & \cos\theta \\ 1 & \cos\theta & \sin\theta \\ 1 & 1 & 1 \end{vmatrix}$

$R_1 \to R_1 - R_3$ and $R_2 \to R_2 - R_3$:

$D(\theta) = \begin{vmatrix} 0 & \sin\theta - 1 & \cos\theta - 1 \\ 0 & \cos\theta - 1 & \sin\theta - 1 \\ 1 & 1 & 1 \end{vmatrix}$

$D(\theta) = (\sin\theta - 1)^2 - (\cos\theta - 1)^2$

$= (\sin\theta - \cos\theta)(\sin\theta + \cos\theta - 2)$

Since $\sin\theta + \cos\theta = \sqrt{2}\sin\left(\theta + \dfrac{\pi}{4}\right) \leq \sqrt{2} < 2,$

the term $(\sin\theta + \cos\theta - 2) \neq 0$ for all real θ.

Thus, $D(\theta) = 0 \iff \sin\theta = \cos\theta$

$\iff \tan\theta = 1.$

In $[0, 2\pi]$, $\tan\theta = 1$ $\theta = \dfrac{\pi}{4}$ and $\theta = \dfrac{5\pi}{4}$.

Hence, option (A) is correct.

$D(\theta) = 0$, $\sin\theta = \cos\theta = k$, where $k = \pm\dfrac{1}{\sqrt{2}}$.

The first and third equations become:

$x + k(y + z) = 0 \quad \text{and} \quad x + y + z = 0$

Substituting $y + z = -x$ into the first equation yields $x(1 - k) = 0$.

Since $k \neq 1$, we must have $x = 0$.

Thus, for all non-trivial solutions, $x = 0$ (with $y = -z$).

Hence, option (B) is correct.

For $\theta = \dfrac{\pi}{2}$, $D\left(\dfrac{\pi}{2}\right) = (1 - 0)(1 + 0 - 2) = -1 \neq 0$,

$x = y = z = 0$ (a unique solution).

Hence, option (C) is correct.

For $\theta = \dfrac{3\pi}{4}$, $\tan\left(\dfrac{3\pi}{4}\right) = -1 \neq 1$, so $D\left(\dfrac{3\pi}{4}\right) \neq 0$.

Paragraph

For $\theta \in (0, \pi)$, consider the functions:

$A(\theta) = \sum_{k=1}^{6} \cos(2k\theta)$

$B(\theta) = \sum_{k=1}^{6} \sin(2k\theta)$

$C(\theta) = 1 + 2\sum_{k=1}^{3} \cos((2k - 1)\theta)$

Consider the parametric quadratic equation in x:

$A(\theta)x^2 + B(\theta)x + C(\theta) = 0$

Q9. The number of values of $\theta \in (0, \pi)$ for which the quadratic equation has both roots approaching infinity is:

(A) 4 (B) 5 (C) 6 (D) 0

Ans. B

Sol. $A(\theta) = \dfrac{\sin(6\theta)\cos(7\theta)}{\sin\theta}$

$B(\theta) = \dfrac{\sin(6\theta)\sin(7\theta)}{\sin\theta}$

$C(\theta) = 1 + \dfrac{\sin(6\theta)}{\sin\theta}$

Here, $A(\theta) = 0$ and $B(\theta) = 0$ simultaneously when $\sin(6\theta) = 0$ for $\theta \in (0, \pi)$.

$\sin(6\theta) = 0 \implies 6\theta = k\pi \implies \theta = \dfrac{k\pi}{6}$ for $k \in \{1, 2, 3, 4, 5\}$.

For all these 5 values of θ, $C(\theta) = 1 + 0 = 1 \neq 0$.

Thus, both roots tend to infinity for exactly 5 values of θ.

Q10. Let Ω be the set of all values of $\theta \in (0, \pi)$ for which the quadratic equation has exactly one root approaching infinity. For each $\theta \in \Omega$, let $x(\theta)$ denote the finite root of the equation. The value of $\sum_{\theta \in \Omega} x(\theta)$ is:

(A) -2 (B) 0 (C) 2 (D) -1

Ans. A

Sol. $A(\theta) = \dfrac{\sin(6\theta)\cos(7\theta)}{\sin\theta}$

$B(\theta) = \dfrac{\sin(6\theta)\sin(7\theta)}{\sin\theta}$

$C(\theta) = 1 + \dfrac{\sin(6\theta)}{\sin\theta}$

$A(\theta) = 0$ and $B(\theta) \neq 0$.

$A(\theta) = 0 \implies \cos(7\theta) = 0$ and $\sin(6\theta) \neq 0$.

$\cos(7\theta) = 0 \implies 7\theta = \dfrac{(2m+1)\pi}{2}$

$\implies \theta_m = \dfrac{(2m+1)\pi}{14}$ for $m \in \{0,1,2,3,4,5,6\}$.

For $m = 3$, $\theta = \dfrac{\pi}{2}$, which gives $\sin(6\theta) = \sin(3\pi) = 0$ (excluded).

Thus, $\Omega = \left\{\theta_m = \dfrac{(2m+1)\pi}{14} : m \in \{0,1,2,4,5,6\}\right\}$.

For $\theta \in \Omega$, the finite root is:

$x(\theta) = -\dfrac{C(\theta)}{B(\theta)} = -\dfrac{1 + \frac{\sin(6\theta)}{\sin\theta}}{\frac{\sin(6\theta)\sin(7\theta)}{\sin\theta}}$

$= -\dfrac{\sin\theta + \sin(6\theta)}{\sin(6\theta)\sin(7\theta)}$

$7\theta_m = \dfrac{(2m+1)\pi}{2}$, $\sin(7\theta_m) = (-1)^m$ and $\sin(6\theta_m) = \sin(7\theta_m - \theta_m) = (-1)^m\cos\theta_m$.

$x(\theta_m) = -\dfrac{\sin\theta_m + (-1)^m\cos\theta_m}{(-1)^m\cos\theta_m \cdot (-1)^m}$

$= -\tan\theta_m - (-1)^m$

$m \in \{0,1,2,4,5,6\}$:

$\pi/2, \sum\limits_{m\in\{0,1,2,4,5,6\}} \tan\theta_m = 0.$

Therefore:

$\sum\limits_{\theta\in\Omega} x(\theta) = -\sum\limits_{m\in\{0,1,2,4,5,6\}} (-1)^m$

$= -(1 - 1 + 1 + 1 - 1 + 1) = -2$

Q1. Consider the polynomial $P(x) = c_5x^5 + c_4x^4 + c_3x^3 + c_2x^2 + c_1x + c_0$, where the coefficients are defined as:

$c_5 = \log_2(\log_3 81) + \log_3(\log_2 8)$

$c_4 = \log_2 3 \cdot \log_3 4 - \log_5 25$

$c_3 = \log_3 18 - \log_3 2 - \log_2 4$

$c_2 = 4^{\log_2 3} - 3^{\log_3 9}$

$c_1 = \log_{\frac{1}{3}}\left(\dfrac{1}{27}\right) + \log_4 2$

$c_0 = \log_2 5 \cdot \log_5 8 - \log_3 3$

If N represents the total number of non-real complex roots of the equation $P(x) = 0$, then find the value of N_____.

Ans. 4

Sol. $c_5 = \log_2(4) + \log_3(3) = 2 + 1 = 3 > 0$

$c_4 = \left(\frac{\ln 3}{\ln 2}\right) \cdot \left(\frac{2\ln 2}{\ln 3}\right) - 2 = 2 - 2 = 0$

$c_3 = \log_3\left(\frac{18}{2}\right) - 2 = \log_3 9 - 2 = 2 - 2 = 0$

$c_2 = (2^{\log_2 3})^2 - 9 = 3^2 - 9 = 0$

$c_1 = 3 + \frac{1}{2} = \frac{7}{2} > 0$

$c_0 = \log_2 8 - 1 = 3 - 1 = 2 > 0$

$P(x) = 3x^5 + \frac{7}{2}x + 2$

For $P(x) = 3x^5 + \frac{7}{2}x + 2$

The number of sign variations $V_+ = 0$, so $P(x) = 0$ has no positive real roots.

For $P(-x) = -3x^5 - \frac{7}{2}x + 2$

The number of sign variations $V = 1$, so $P(x) = 0$ has at most 1 negative real root.

$P(0) = 2 > 0$ and $\lim_{x \to -\infty} P(x) = -\infty$,

by the Intermediate Value Theorem, $P(x) = 0$ has at least 1 real root.

Therefore, $P(x) = 0$ has exactly 1 real root.

Since the total degree of $P(x)$ is 5, the total number of roots in $\mathbb{C}$ is 5.

The number of non-real complex roots is:

$N = 5 - 1 = 4$

Q2. Let A and B be two 3×3 real matrices such that A is symmetric and B is skew-symmetric. Define the matrix $C = AB - BA$. Suppose $\det(C) = 0$ and $\mathrm{Tr}(C^2) = 50$. If the homogeneous system of linear equations $(C - kI_3)X = 0$ has a non-trivial solution for some $k \in \mathbb{R}$, where I_3 is the 3×3 identity matrix, then the maximum possible value of $|k|$ is ____.

Ans. 5

Sol. $C^T = (AB - BA)^T = B^T A^T - A^T B^T$

$= (-B)A - A(-B) = -BA + AB$

$= AB - BA = C$

Thus, C is a real symmetric matrix, which implies all its eigenvalues are real.

$\mathrm{Tr}(C) = \mathrm{Tr}(AB - BA) = \mathrm{Tr}(AB) - \mathrm{Tr}(BA) = 0$

Let $\lambda_1, \lambda_2, \lambda_3$ be the eigenvalues of C.

1. $\mathrm{Tr}(C) = \lambda_1 + \lambda_2 + \lambda_3 = 0$

2. $\det(C) = \lambda_1\lambda_2\lambda_3 = 0$

Since $\det(C) = 0$, at least one eigenvalue must be zero. Let $\lambda_3 = 0$.

Then $\lambda_1 + \lambda_2 = 0 \implies \lambda_2 = -\lambda_1$.

Now, the trace of C^2 is the sum of squares of the eigenvalues of C:

$\text{Tr}(C^2) = \lambda_1^2 + \lambda_2^2 + \lambda_3^2$

$= \lambda_1^2 + (-\lambda_1)^2 + 0 = 2\lambda_1^2$

Given $\text{Tr}(C^2) = 50$:

$2\lambda_1^2 = 50 \implies \lambda_1^2 = 25 \implies \lambda_1 = 5,\ \lambda_2 = -5$

The eigenvalues of C are $5, -5, 0$.

For the system $(C - kI_3)X = 0$ to have a non-trivial solution, we must have:

$\det(C - kI_3) = 0$

This means k must be an eigenvalue of C.

Thus, $k \in \{5, -5, 0\}$.

Therefore, the maximum possible value of $|k|$ is 5.

Q3. Let A and B be two 3×3 real matrices such that A is symmetric and B is skew-symmetric. Suppose $\text{Tr}(A^2) = 1$ and $\text{Tr}(B^2) = -1$. For $x \in \mathbb{R}$, consider the function $f(x) = \dfrac{2x^2 - x + \text{Tr}((A + xB)^2)}{2x^2 + x + \text{Tr}((A + xB)^2)}$

If the range of $f(x)$ is $[m, M]$, then find the value of $\dfrac{M}{m}$_____.

Ans. 9

Sol. $\text{Tr}(AB) = \text{Tr}((AB)^T) = \text{Tr}(B^T A^T)$

$\implies 2\text{Tr}(AB) = 0 \implies \text{Tr}(AB) = 0$

Now, expanding $\text{Tr}((A + xB)^2)$:

$\text{Tr}((A + xB)^2) = \text{Tr}(A^2 + x(AB + BA) + x^2 B^2)$

$= \text{Tr}(A^2) + 2x\text{Tr}(AB) + x^2\text{Tr}(B^2)$

$= 1 + 2x(0) + x^2(-1) = 1 - x^2$

Substituting this into $f(x)$:

$f(x) = \dfrac{2x^2 - x + (1 - x^2)}{2x^2 + x + (1 - x^2)} = \dfrac{x^2 - x + 1}{x^2 + x + 1}$

$y(x^2 + x + 1) = x^2 - x + 1$

$\implies (y - 1)x^2 + (y + 1)x + (y - 1) = 0$

$D \geq 0$:

$(y + 1)^2 - 4(y - 1)^2 \geq 0$

$\implies ((y + 1) - 2(y - 1))((y + 1) + 2(y - 1)) \geq 0$

$\implies (3 - y)(3y - 1) \geq 0 \implies \dfrac{1}{3} \leq y \leq 3$

Thus, $m = \dfrac{1}{3}$ and $M = 3$.

Hence, $\dfrac{M}{m} = \dfrac{3}{1/3} = 9$.

Q4. Consider the trigonometric sequence defined by angles in arithmetic progression. Let S be the value of the finite sum $S = \displaystyle\sum_{k=1}^{6} \frac{1}{\sum_{j=1}^{k} \sin\left(\frac{j\pi}{6}\right)}$

Find the value of S _____.

Ans. 4

Sol. $D_k = \sum_{j=1}^{k} \sin(j\theta) = \dfrac{\sin\left(\frac{k\theta}{2}\right)\sin\left(\frac{(k+1)\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}$

$\dfrac{1}{D_k} = \dfrac{\sin\left(\frac{\theta}{2}\right)}{\sin\left(\frac{k\theta}{2}\right)\sin\left(\frac{(k+1)\theta}{2}\right)}$

$\dfrac{1}{D_k} = \cot\left(\dfrac{k\theta}{2}\right) - \cot\left(\dfrac{(k+1)\theta}{2}\right)$

$S = \sum_{k=1}^{6} \dfrac{1}{D_k} = \cot\left(\dfrac{\theta}{2}\right) - \cot\left(\dfrac{7\theta}{2}\right)$

$\theta = \dfrac{\pi}{6}$:

$\dfrac{\theta}{2} = \dfrac{\pi}{12}$ and $\dfrac{7\theta}{2} = \dfrac{7\pi}{12} = \dfrac{\pi}{2} + \dfrac{\pi}{12}$

$S = \cot\left(\dfrac{\pi}{12}\right) - \left(-\tan\left(\dfrac{\pi}{12}\right)\right)$

$= \cot\left(\dfrac{\pi}{12}\right) + \tan\left(\dfrac{\pi}{12}\right)$

$S = 2\csc\left(2 \cdot \dfrac{\pi}{12}\right) = 2\csc\left(\dfrac{\pi}{6}\right)$

$= 2 \times 2 = 4$

Q1. The number of real values of x in the interval $[0, 3\pi]$ satisfying the equation $4^{\sin^2 x} + 4^{\cos^2 x} = 4$ is equal to _____.

Ans. 6

Sol. $4^{\sin^2 x} + 4^{\cos^2 x} = 4$

$4^{\sin^2 x} + 4^{1-\sin^2 x} = 4$

Let $t = 4^{\sin^2 x}$.

$t + \dfrac{4}{t} = 4$

$t^2 - 4t + 4 = 0$

$(t-2)^2 = 0$

So, $t = 2$.

$4^{\sin^2 x} = 2$

$2^{2\sin^2 x} = 2^1$

Equating the exponents:

$2\sin^2 x = 1$

$\sin^2 x = \dfrac{1}{2}$

In $[0, 2\pi]$:

- $\sin x = \dfrac{1}{\sqrt{2}}$ gives $x = \dfrac{\pi}{4}, \dfrac{3\pi}{4}$.
- $\sin x = -\dfrac{1}{\sqrt{2}}$ gives $x = \dfrac{5\pi}{4}, \dfrac{7\pi}{4}$.

So, in $[0, 2\pi]$, there are 4 solutions.

In $(2\pi, 3\pi]$:

$\sin x = \dfrac{1}{\sqrt{2}}$ gives $x = 2\pi + \dfrac{\pi}{4} = \dfrac{9\pi}{4}$ and $x = 2\pi + \dfrac{3\pi}{4} = \dfrac{11\pi}{4}$.

$\sin x = -\dfrac{1}{\sqrt{2}}$ has no solutions in $(2\pi, 3\pi]$

Total number of real solutions = $2 + 2 + 2 = 6$.

Q2. Let $A(x) = \begin{bmatrix} x-1 & 1 \\ x^2 - 5x + 6 & 3 - x \end{bmatrix}$ for $x \in \mathrm{R}$, and let $B(x) = A(x) - I_2$, where I_2 is the 2×2 identity matrix.

Let α_1 and α_2 be the roots of x for which $B(x)$ is a nilpotent matrix, where $\alpha_1 \in \mathrm{Z}$ is the integral root and $\alpha_2 \notin \mathrm{Z}$ is the rational non-integral root.

Let β_1 and β_2 be the roots of x for which $B(x)$ is an involutory matrix, where $\beta_1 \in \mathrm{Z}$ is the integral root and $\beta_2 \notin \mathrm{Z}$ is the rational non-integral root.

Let γ_1 and γ_2 be the real roots of the equation $\det\left((B(x))^3 + 8I_2\right) = 0$.

Find the value of the expression: $V = (\alpha_1 + \beta_1)^2 + (\alpha_2 + \beta_2)(\gamma_1 + \gamma_2) + \gamma_1\gamma_2$

Ans. 46

Sol. $B(x) = \begin{bmatrix} x-2 & 1 \\ x^2 - 5x + 6 & 2 - x \end{bmatrix}$

$\mathrm{tr}(B(x)) = (x-2) + (2-x) = 0$

$\det(B(x)) = (x-2)(2-x) - 1.(x^2 - 5x + 6)$

$= -2x^2 + 9x - 10$

$B(x)^2 - \mathrm{tr}(B(x))B(x) + \det(B(x))I_2 = O$

$B(x)^2 = -\det(B(x))I_2$

$= (2x^2 - 9x + 10)I_2$

Let $Q(x) = 2x^2 - 9x + 10$.

Thus, $B(x)^2 = Q(x)I_2$.

Case 1: $B(x)$ is Nilpotent

$B(x)$ is nilpotent $\iff B(x)^2 = O \iff Q(x) = 0$.

$2x^2 - 9x + 10 = 0 \implies (2x - 5)(x - 2) = 0$

The roots are $x = 2$ and $x = \dfrac{5}{2}$.

Integral root: $\alpha_1 = 2$

Rational non-integral root: $\alpha_2 = \dfrac{5}{2}$

Case 2: $B(x)$ is Involutory

$B(x)$ is involutory $\iff B(x)^2 = I_2 \iff Q(x) = 1$.

$2x^2 - 9x + 10 = 1$

$\implies 2x^2 - 9x + 9 = 0$

$\implies (2x-3)(x-3)=0$

The roots are $x=3$ and $x=\dfrac{3}{2}$.

Integral root: $\beta_1=3$

Rational non-integral root: $\beta_2=\dfrac{3}{2}$

Case 3: Real roots of $\det\left((B(x))^3+8I_2\right)=0$

Using $B(x)^2=Q(x)I_2$, we have:

$(B(x))^3=B(x)^2.B(x)=Q(x)B(x)$

Thus, $(B(x))^3+8I_2=Q(x)B(x)+8I_2$.

For any 2×2 matrix M with $\text{tr}(M)=0$ and $\det(M)=-Q(x)$,

we have $\det(kM+cI_2)=c^2+k^2\det(M)=c^2-k^2Q(x)$.

Here $k=Q(x)$ and $c=8$, so:

$\det(Q(x)B(x)+8I_2)$

$=8^2-(Q(x))^2(-Q(x))=64-(Q(x))^3$

Setting the determinant to zero:

$64-(Q(x))^3=0 \implies Q(x)=4$

$2x^2-9x+10=4 \implies 2x^2-9x+6=0$

The roots γ_1,γ_2 are real (as $D=81-48=33>0$).

$\gamma_1+\gamma_2=\dfrac{9}{2},\ \gamma_1\gamma_2=\dfrac{6}{2}=3$

$\alpha_1+\beta_1=2+3=5$

$\alpha_2+\beta_2=\dfrac{5}{2}+\dfrac{3}{2}=4$

$V=(5)^2+(4)\left(\dfrac{9}{2}\right)+3$

$=25+18+3=46$

Q3. Let $\{x_k\}_{k=1}^{\infty}$ be a sequence of positive real numbers satisfying $\displaystyle\sum_{k=1}^{\infty}\left(x_k+\frac{k^2}{(k+1)!^2\,x_k}\right)\le 2$.

For each $k\ge 1$, let $y_k=\dfrac{1}{(k+1)!}$ and let H_k denote the Harmonic Mean of x_k and y_k. Then the value of the infinite sum $\displaystyle\sum_{k=1}^{\infty}\frac{H_k}{x_k\,y_k\,(k+2)!}$ is equal to _____.

Ans. 1

Sol. $x_k+\dfrac{k^2}{(k+1)!^2\,x_k}\ge 2\sqrt{x_k\cdot\dfrac{k^2}{(k+1)!^2\,x_k}}$

$=\dfrac{2k}{(k+1)!}.$

$\displaystyle\sum_{k=1}^{\infty}\frac{2k}{(k+1)!}=2\sum_{k=1}^{\infty}\left(\frac{(k+1)-1}{(k+1)!}\right)$

$=2\displaystyle\sum_{k=1}^{\infty}\left(\frac{1}{k!}-\frac{1}{(k+1)!}\right)=2\,(1-0)=2.$

$\displaystyle\sum_{k=1}^{\infty}\left(x_k+\frac{k^2}{(k+1)!^2\,x_k}\right)\le 2$, equality must hold in the AM-GM inequality for every $k\ge 1$.

$$x_k = \frac{k^2}{(k+1)!^2\, x_k} \implies x_k = \frac{k}{(k+1)!}.$$

$$y_k = \frac{1}{(k+1)!}$$

$$H_k = \frac{2x_k y_k}{x_k + y_k} \implies \frac{H_k}{x_k y_k} = \frac{2}{x_k + y_k}.$$

$$x_k + y_k = \frac{k}{(k+1)!} + \frac{1}{(k+1)!}$$

$$= \frac{k+1}{(k+1)!} = \frac{1}{k!}:$$

$$\frac{H_k}{x_k y_k} = 2 \cdot k!.$$

$$S = \sum_{k=1}^{\infty} \frac{H_k}{x_k\, y_k\, (k+2)!} = \sum_{k=1}^{\infty} \frac{2 \cdot k!}{(k+2)!}$$

$$= \sum_{k=1}^{\infty} \frac{2}{(k+1)(k+2)}.$$

$$S = 2 \sum_{k=1}^{\infty} \left(\frac{1}{k+1} - \frac{1}{k+2} \right)$$

$$= 2 \left(\frac{1}{2} \right) = 1.$$